

■ Notes on Firmware

1. Definition

- Firmware = Low-level software stored in non-volatile memory (Flash/ROM/EEPROM) that directly controls hardware.
- Acts as a bridge between raw hardware and higher-level software (OS, apps, user code).

2. Key Properties

- Hardware-specific (tied to one device).
- Runs directly on the ISA (Instruction Set Architecture) of the processor.
- Stored in Flash/ROM → survives power loss.
- Provides fundamental hardware control.

3. Firmware vs. Software

- Firmware: Controls hardware, hardware-specific, in Flash/ROM.
- Software: Runs on OS, portable, stored in SSD/HDD.

4. Memory in Microcontrollers

- Flash Memory → Stores firmware/program (non-volatile).
- SRAM → Temporary variables during execution (volatile).
- EEPROM → Small non-volatile memory for configs (WiFi creds, calibration).

5. Uploading vs. Burning Firmware

- Uploading Code → Compiled sketch written into Flash via bootloader (Arduino IDE upload).
- Burning Firmware → Replacing bootloader/system firmware via external programmer (USBasp, etc.).

6. OTA (Over-The-Air) Updates

- Updating firmware wirelessly (WiFi/Bluetooth/cellular).
- Process: Download → Store → Replace old firmware → Restart.
- Examples: ESP32 OTA, smartphone updates, Tesla cars, DJI drones.
- Advantage: No physical connection needed, crucial for IoT/remote devices.

7. Instruction Set & Firmware

- Firmware executes as machine code for the ISA of the processor.
- Example: Arduino Uno → AVR ISA, ESP32 → Xtensa/RISC-V ISA, PC → x86 ISA.
- All firmware is ISA code, but not all ISA code is firmware (apps/games also ISA code but not firmware).

8. BIOS as Firmware

- BIOS/UEFI = Firmware for PCs.
- Written in C + Assembly, compiled into x86 machine code.
- Stored in ROM/Flash on motherboard.
- Role: Hardware initialization, boot device detection, load OS.

9. Real-Life Examples of Firmware

- Arduino sketches (after flashing).
- Arduino bootloader.
- BIOS/UEFI in PCs.
- Router firmware.
- Drone flight controller firmware (Betaflight, Ardupilot).
- Smartphone baseband firmware.

■ Final Takeaways

- Firmware = Hardware-bound software in Flash/ROM that runs on the processor's ISA to control hardware.
- Arduino sketches are firmware once flashed.
- BIOS is firmware written in C/Assembly, compiled to ISA instructions, stored in ROM.
- OTA = Wireless firmware update.

■ Abbreviations

- ISA → Instruction Set Architecture
- ROM → Read Only Memory
- EEPROM → Electrically Erasable Programmable Read Only Memory
- SRAM → Static Random Access Memory
- BIOS → Basic Input/Output System
- UEFI → Unified Extensible Firmware Interface
- OTA → Over-The-Air
- USBasp → USB In-System Programmer (common AVR programmer)